

Magnetoelectric effects in superconductor/ferromagnet bilayers

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We demonstrate that the hybrid structures consisting of a superconducting layer with an adjacent spin-textured ferromagnet demonstrate the variety of equilibrium magnetoelectric effects originating from coupling between conduction electron spin and superconducting current. By deriving and solving the generalized Usadel equation which takes into account the spin-filtering effect we find that a supercurrent generates spin polarization in the superconducting film which is non-coplanar with the local ferromagnetic moment. The inverse magnetoelectric effect in such structures is shown to result in the spontaneous phase difference across the magnetic topological defects such as a domain wall and helical spin texture. The possibility to obtain dissipationless spin torques and detect domain wall motion through the superconducting phase difference are discussed.

I. INTRODUCTION

Magnetoelectric effects resulting from the intrinsic spin-orbital coupling (SOC) have been studied quite intensively in different conducting materials with inversion asymmetry^{1–9}. The direct magnetoelectric effect which is the generation of spin polarization by local electric fields^{1–4} has been experimentally observed by the optical probes^{5–7,10}, electronic resonance techniques¹¹ and direct electrical measurements^{12,13}. Recently, it has become the topic of great interest in connection with magnetic memory applications based on the spin-orbit torque mechanism of magnetization switching^{14–19} and domain wall motion²⁰. The inverse magnetoelectric effect or the spin galvanic effect is the generation of charge current due to the nonequilibrium spin polarization¹. It has been observed experimentally in semiconductors^{8,9} and normal metals²¹.

The equilibrium counterparts of magnetoelectric effects discussed above exist in superconducting materials resulting from the coupling between supercurrent and various magnetic degrees of freedom. These can be either magnetic moments of conductivity electrons forming spin-triplet Cooper pairs or the localized spins responsible for magnetically ordered state. Up to now the direct magneto-electric effect was reported for superconducting systems in the presence of either intrinsic^{22–27} or extrinsic SOC²⁸. A number of works predicted an anomalous Josephson effect, which can be viewed²⁹ as an inverse magnetoelectric effect, specific for Josephson junctions. Its essence is that a spontaneous phase difference intermediate between 0 and π appears in the ground state of the junction. It was proposed for interlayers with SOC under the applied magnetic field^{29–43} and for non-coplanar magnetic interlayers^{39,42–51}. The unified theory of anomalous Josephson effect which combines the presence of SOC and magnetic texture has been developed recently⁵². This effect has been recently observed in the Josephson junctions through a quantum dot⁵³ and Bi₂Se₃

interlayer^{54,55}.

Given the analogy between intrinsic SOC and the SU(2) gauge field⁵⁶ which can be induced by the spatial rotations of magnetization in ferromagnets (F) it is natural to expect that equilibrium magnetoelectric effect should exist also in spin-textured S/F hybrid structures. However in contrast to superconducting systems with SOC the direct magnetoelectric effect in S/F hybrids with spin-singlet pairing has not been obtained in the previous theoretical works. This is despite the possibility to produce spin-triplet Cooper pairs in spin-textured S/F systems is well known^{57,58} and proposed to have a lot of future applications in spintronics and superconducting electronics^{59,60}. The reason for that roots in the limitations of the quasiclassical theory which has been widely used for the study of such systems and which neglects the difference in the Fermi velocities and the density of states in spin-up and spin-down subbands. Although such approximation is justified for the description of weak ferromagnets like dilute magnetic alloys it generically misses the anomalous Josephson effect⁴³ and as we show below, the direct magnetoelectric effects as well.

To go beyond the limitations of quasiclassical theory we consider its minimal extension for a bilayer S/F system consisting of thin superconducting film separated by the tunnel barrier from the strong spin-textured ferromagnet. We demonstrate that this system features both the direct and inverse magnetoelectric effects. The latter produces the phase-inhomogeneous superconducting ground state and can be used for electrical detection of the domain wall motion and its chirality. The former leads to the generation of non-collinear spin density in the superconducting film and can be used for generating the spin torque acting on the spin texture in the adjacent ferromagnet.

Equilibrium supercurrent-induced spin torques were discussed previously in Josephson junctions through the single-domain magnets^{33,61–64}, layered systems^{47,65–67}, in ferromagnetic⁶⁸ and spin-triplet superconductors⁶⁹.

Supercurrent-driven domain wall motion in the Josephson junction through strong ferromagnet was considered in⁷⁰. Here we analyse this effect for S/F bilayers with singlet superconductors and elucidate its connection with the direct magnetoelectric effect. The current-induced spin polarization in superconducting field is found to have a component perpendicular to the local magnetization in the adjacent ferromagnet. Therefore it gives rise to the spin torque acting on the ferromagnet texture $\mathbf{M}(\mathbf{r})$. The torque that we have found has two contributions. The first one is similar to the usual adiabatic spin-transfer torque in normal^{71–75} and superconducting systems⁷⁰. The second part of the torque is intimately connected to the local chirality of the magnetization texture and is analogous neither to the adiabatic spin transfer torque, nor to a non-adiabatic (anti-damping) torque⁷⁶.

The paper is organized as follows. In Sec. II we discuss essential ingredients for obtaining magnetoelectric effects in S/F bilayers and develop a theoretical approach for treating superconductivity in such structures. In Sec. III the spontaneous phase gradients in the ground state of the S/F bilayer containing a magnetization texture are obtained and possible applications of this effect for electrical detection of the DW presence, motion and chirality are discussed. Sec. IV is devoted to the calculation and discussion of the supercurrent-induced spin polarization in the superconductor and the resulting torque acting on the DW.

II. GENERALIZED USADEL EQUATION

We consider a thin (the thickness d along the z direction is much smaller than the superconducting coherence length) superconducting film in a contact with a ferromagnet. The sketch of different system configurations, which we consider, is presented in Fig. 1. The interface is assumed to be in the $z = 0$ plane. Below we denote $\mathbf{r} = (x, y)$ to be 2D coordinate vector in plane of the film. The interface is low-transparent in order to prevent the complete suppression of superconductivity by the exchange field of the ferromagnet.

It is widely accepted in the literature that if the thickness of the S film d is smaller than the superconducting coherence length ξ_S , the magnetic proximity effect, that is the influence of the adjacent ferromagnet with the magnetization $\mathbf{M}(\mathbf{r})$ on the S film can be described by adding the effective exchange field $\mathbf{h}_{eff}(\mathbf{r}) \propto \mathbf{M}(\mathbf{r})$ to the quasiclassical Eilenberger or Usadel equation, which is used for treating the superconductor. This was reported as for metallic⁵⁷, so as for insulating^{77,78} ferromagnets. While for the ferromagnetic insulators the magnetic proximity effect is not so simple and in general is not reduced to the effective exchange only^{78–80}, the other terms (which can be viewed as additional magnetic impurities in the superconductor) are omitted in Eq. (1) for simplicity because they do not influence the discussed below effects

qualitatively. Hence we consider the Usadel equation in the form

$$-iD\nabla(\check{g}_S\nabla\check{g}_S) + [\check{\Lambda} - \check{\Delta}(\mathbf{r}), \check{g}_S] = 0, \quad (1)$$

where $\check{g}_S \equiv \check{g}_S(\mathbf{r}, \omega)$ is the momentum-averaged quasiclassical Green's function and ω is the Matsubara frequency, respectively. The matrix gap function is $\check{\Delta}(\mathbf{r}) = \Delta(\mathbf{r})\hat{\tau}_+ - \Delta^*(\mathbf{r})\hat{\tau}_-$ with $\hat{\tau}_\pm = (\hat{\tau}_x \pm i\hat{\tau}_y)/2$ and the diagonal spin-dependent potential term is given by

$$\check{\Lambda} = \check{\Lambda}_0 \equiv \hat{\tau}_z[i\omega + \boldsymbol{\sigma}\mathbf{h}_{eff}(\mathbf{r})] \quad (2)$$

We denote $\hat{\sigma}_i$ and $\hat{\tau}_i$ to be the Pauli matrices in spin and particle-hole spaces, respectively.

Although Eq.(1) describes the formation of spin-triplet superconducting correlations in the spatially-inhomogeneous field $\mathbf{h}_{eff}(\mathbf{r})$, it completely misses the magnetoelectric effects, which can be understood from the following argument. In general, the supercurrent $\mathbf{j}$ flowing through the system is a function of effective exchange field $\mathbf{h}_{eff} \propto \mathbf{M}$ and the superconducting phase gradient $\nabla\varphi(\mathbf{r})$. The time reversal symmetry dictates that $\mathbf{j}(\nabla\varphi, \mathbf{M}) = -\mathbf{j}(-\nabla\varphi, -\mathbf{M})$. It was shown⁴³ that for a system described by Eq. (1) the additional quasiclassical symmetry holds $\mathbf{j}(\mathbf{M}) = \mathbf{j}(-\mathbf{M})$. Combining this with the time-reversal symmetry we obtain $\mathbf{j}(\nabla\varphi) = -\mathbf{j}(-\nabla\varphi)$. Consequently, the anomalous supercurrent at zero $\nabla\varphi$ and the phase-inhomogeneous superconducting ground states are not allowed. In order to allow for the magnetoelectric effects two conditions should be satisfied simultaneously: (i) the magnetization $\mathbf{M}$ of the ferromagnet should be noncoplanar and (ii) the magnetization of the ferromagnet should be treated beyond the quasiclassical approximation Eq. (1) in order to violate the quasiclassical symmetry $\mathbf{j}(\mathbf{M}) = \mathbf{j}(-\mathbf{M})$ ⁴³.

Below we demonstrate that the minimal necessary generalization of the Eilenberger equation allowing for describing magnetoelectric effects is to include the "spin-dependent depairing" term which modifies the diagonal potential in Eq.(1)

$$\check{\Lambda} = \check{\Lambda}_0 + i\Gamma\text{sgn}\omega(\hat{\tau}_z + \mathbf{P}\boldsymbol{\sigma}) \quad (3)$$

Qualitatively the last term in Eq.(3) describes the suppression of superconductivity in the film due to spin-dependent tunneling of electrons forming Coopers pairs into the adjacent normal ferromagnet and Γ is the effective depairing parameter. The polarization $\mathbf{P}$ describes the efficiency and quantization axis of the spin-filter which acts on the electrons during the tunnelling between the superconductor and adjacent layer, ferromagnetic or normal. If the effective exchange and the spin filtering tunneling are provided by the same ferromagnet [Fig.1(a)], the direction of $\mathbf{P}$ coincides with that of the effective exchange field $\mathbf{P} \parallel \mathbf{h}_{eff}$, although in principle they can differ e.g. due to the small-scale magnetic inhomogeneity at the interface layer. The condition $\mathbf{h}_{eff} \nparallel \mathbf{P}$ can also be reached in a hybrid structure

like the sketched in Fig. 1(b), where a weak ferromagnet inducing h_{eff} and a spin-filtering interface are spatially separated by the superconducting film.

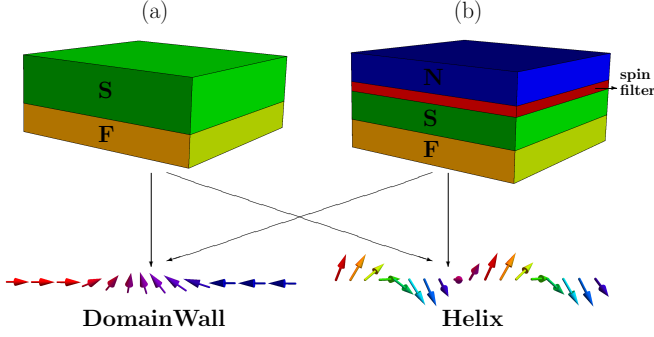


FIG. 1. Sketch of systems under consideration. The left figure represents a possible realization of S/F bilayer consisting of a conventional superconducting film and a strong ferromagnet. The right figure is a possible realization of a S/F bilayer with a weak ferromagnet and a spin-filtering interface. The spin filtering interface and the ferromagnet are separated by the superconducting film in order to allow for misaligned magnetizations of these magnetic layers. The considered magnetic texture configurations are shown in the bottom of the figure.

Below by an example of a S/F bilayer with spin-filtering interface we elucidate the physical origin of the spin-filtering term in Eq.(3). The system includes a singlet superconductor and a weak ferromagnet with the exchange field $h \ll \varepsilon_F$. We treat the whole bilayer as an effective ferromagnetic superconductor with the order parameter Δ and the exchange field $\mathbf{h}_{eff}(\mathbf{r})$. It is assumed that one of the interfaces of the bilayer is spin-filtering, that is it has different conductances for spin up and spin down electrons. We prefer to make the upper interface of the bilayer spin-filtering in order to spatially separate it from the ferromagnet to have a possibility to misalign the magnetization vectors of the ferromagnet and the spin-filtering interface. The sketch of the system is presented in Fig. 1(b). In this case the bilayer is described by the standard Usadel equation:

$$-D[\nabla_{\mathbf{r}}(\tilde{g}\nabla_{\mathbf{r}}\tilde{g}) + \partial_z(\tilde{g}\partial_z\tilde{g})] + [\omega\tau_z - i\mathbf{h}_{eff}(\mathbf{r})\boldsymbol{\sigma}\tau_z + i\tilde{\Delta}(\mathbf{r}), \tilde{g}] = 0, \quad (4)$$

which should be supplemented by the appropriate boundary conditions. At the impenetrable surface $z = 0$ it takes the form⁸¹:

$$\tilde{g}\partial_z\tilde{g} = 0 \quad (5)$$

and at the spin-filtering interface $z = d$ it takes the form of the generalized Kuprianov-Lukichev boundary conditions⁸¹, that include spin-polarized tunnelling at the SF interfaces^{80,82-84}:

$$\gamma\tilde{g}\partial_z\tilde{g} = [\tilde{g}, \text{sgn}\omega\tilde{T}\tau_z\tilde{T}^\dagger], \quad (6)$$

where $\tilde{T} = t\sigma_0\tau_0 + u\hat{P}\boldsymbol{\sigma}$ with $t = \sqrt{(1 + \sqrt{1 - P^2})/2}$, $u =$

$\sqrt{(1 - \sqrt{1 - P^2})/2}$ and $\hat{P} = \mathbf{P}/P$. γ is the parameter describing the barrier strength.

Integrating Eq. (1) from $z = 0$ to $z = d$ with the boundary conditions Eqs. (5)-(6) and taking into account that for $d < \xi_S$ we can consider $\tilde{g}$ as spatially constant in the z -direction, we obtain the following effective Usadel equation for the bilayer:

$$-D\nabla(\tilde{g}\nabla\tilde{g}) + [(\omega + \text{sgn}\omega\Gamma)\tau_z - i\mathbf{h}_{eff}\boldsymbol{\sigma}\tau_z + \text{sgn}\omega\Gamma\mathbf{P}\boldsymbol{\sigma} + i\tilde{\Delta}(\mathbf{r}), \tilde{g}] = 0, \quad (7)$$

where we denote $\Gamma = D/\gamma d$.

In fact, Eq. (7) can be obtained for a wide class of systems containing a thin superconducting film with $d \lesssim \xi_S$. For example, the superconducting film in the S/F bilayer with strong ferromagnet [Fig. 1(a)] is described by the same equation, which can be derived making use of the general boundary conditions for spin-active interfaces⁷⁸⁻⁸⁰. In this case as the effective exchange field, so as the $\mathbf{P}$ -term come from the interface properties: while the exchange is a consequence of the spin-mixing⁷⁷ upon the quasiparticle reflection at the S/F interface, the polarization term results from the spin-dependent tunneling through the same interface. In this case Eq. (7) is obtained as a simplified position-independent version of the general result, obtained in Ref. 85 for spatially-inhomogeneous S/F systems. For the discussed model of S/strong ferromagnet bilayer the condition $\mathbf{P} \parallel \mathbf{h}_{eff}$ should be realized as the most probable variant, but, in principle, their directions can differ due to small-scale magnetic inhomogeneities at the interface. The absence of a way to control the mutual orientation of $\mathbf{P}$ and $\mathbf{h}_{eff}$ is a drawback of this model with respect to the system, where the ferromagnet and spin-filtering layer are spatially separated.

III. ANOMALOUS GROUND STATE PHASE SHIFTS IN S/F BILAYERS CONTAINING SPIN TEXTURES

Here we consider the inverse magnetoelectric effect in a S/F bilayer containing a magnetic texture. While the general consideration is valid for an arbitrary texture depending on the only spatial coordinate x , we focus on two particular examples of the magnetic helix and the head-to-head domain wall.

The magnetization texture is described by

$$\mathbf{h} = h(\cos\theta, \sin\theta\cos\delta, \sin\theta\sin\delta), \quad (8)$$

where in general the both angles depend on x -coordinate.

Let's make the spin gauge transform in Eq. (7) in order to work in the reference frame where the quantization axis is aligned with the local magnetization direction: $\tilde{g} = U\tilde{g}_lU^\dagger$ with $U^\dagger\mathbf{h}_{eff}(\mathbf{r})\boldsymbol{\sigma}U = h_{eff}\sigma_z$. Then we

obtain from Eq. (7):

$$-D\hat{\nabla}(\hat{g}_l\hat{\nabla}\hat{g}_l) + \left[(\omega + \text{sgn}\omega\Gamma)\tau_z - ih_{eff}\sigma_z\tau_z + \text{sgn}\omega\Gamma\tilde{\mathbf{P}}\boldsymbol{\sigma} + i\tilde{\Delta}(\mathbf{r}), \hat{g}_l \right] = 0, \quad (9)$$

where $\hat{\nabla} = \nabla + i[M_k^S\sigma_k m_S, \dots]$ is the gauge-covariant derivative with $M_{kj}^S = \text{Tr}[\sigma_k U^\dagger \partial_j U]/2im_S$ and $\tilde{\mathbf{P}}\boldsymbol{\sigma} = U^\dagger \mathbf{P}\boldsymbol{\sigma}U$ is the interface polarization term in the local spin basis. In general $\tilde{\mathbf{P}}$ depends on the x -coordinate even if $\mathbf{P}$ is spatially independent.

The spin rotation is given by

$$\hat{U} = e^{-i\sigma_x\delta/2}e^{-i\sigma_z\theta/2}e^{-i\sigma_y\pi/4}. \quad (10)$$

In this considered case when the magnetization texture only depends on the x -coordinate the gauge field can be written as follows:

$$M_{kx}^S\sigma_k m_S = \mathbf{a}\boldsymbol{\sigma}, \quad (11)$$

where $\mathbf{a} = \frac{1}{2}(\partial_x\theta, (\partial_x\delta)\sin\theta, -(\partial_x\delta)\cos\theta)$ is a vector in spin space. The other components of the spin gauge field M_{ky}^S and M_{kz}^S are zero.

At first we assume that $\mathbf{P}$ is aligned with $\mathbf{h}$ and solve the effective Usadel equation (7) in the S film. For simplicity we consider the linearized version of this equation valid near the critical temperature. The linearized Usadel equation for the anomalous Green's function $\hat{f}_l = f_0\sigma_0 + \mathbf{f}\boldsymbol{\sigma}$ takes the form [we consider $\omega > 0$, for $\omega < 0$ the solutions can be obtained as $f_0(\omega) = f_0(-\omega)$, $\mathbf{f}(\omega) = -\mathbf{f}(-\omega)$]:

$$-\frac{D}{2}\partial_x^2\mathbf{f} + (\omega + \Gamma + 2Da^2)\mathbf{f} + 2D[\mathbf{a} \times \partial_x\mathbf{f}] + \quad (12)$$

$$D[\partial_x\mathbf{a} \times \mathbf{f}] + i\Gamma P[\mathbf{e}_z \times \mathbf{f}] - 2D(\mathbf{a}\mathbf{f})\mathbf{a} - ih_{eff}f_0\mathbf{e}_z = 0, \\ -\frac{D}{2}\partial_x^2f_0 + (\omega + \Gamma)f_0 - ih_{eff}f_z - i\Delta = 0. \quad (13)$$

The general expression for the current reads

$$j_x = \frac{i\pi T}{2e\rho_N} \sum_{\omega>0} \left\{ \text{Tr}_2[\hat{f}_l\partial_x\hat{f}_l - \hat{f}_l\partial_x\hat{f}_l] + 8(\mathbf{f} \times \tilde{\mathbf{f}})\mathbf{a} \right\}, \quad (14)$$

where $\rho_N = 1/(2e^2N_FD)$ is the resistivity of the superconducting film in the normal state.

The electric current can be represented as the sum of the ordinary j_o and anomalous j_a parts. The ordinary and anomalous contributions are given by the first and the second terms in the curly brackets in Eq.(14). The anomalous contribution is defined as the current in the absence of the phase difference $j_a = j(\partial_x\varphi = 0)$. Now our goal is to find the anomalous current, that is to solve Eqs. (12)-(13) at $\partial_x\varphi = 0$. We solve Eqs. (12)-(13) in the approximation of spatially slow magnetic texture with the characteristic length scale $d_W \gg \xi_S$. The solution

up to the leading order in the parameters ξ_S/d_W and $\Gamma P/(\omega + \Gamma)$ takes the form:

$$f_0 = \frac{i\Delta(\omega + \Gamma)}{(\omega + \Gamma)^2 + h_{eff}^2}, \quad (15)$$

$$f_z = \frac{-\Delta h_{eff}}{(\omega + \Gamma)^2 + h_{eff}^2}, \quad (16)$$

$$f_{x,y} = -\frac{Df_z}{(\omega + \Gamma)} \left[\pm\partial_x a_{y,x} - 2a_z a_{x,y} - \frac{i\Gamma P}{(\omega + \Gamma)} (\partial_x a_{x,y} \pm 2a_z a_{y,x}) \right]. \quad (17)$$

The function components $\tilde{f}_i$ can be obtained from the corresponding expressions for f_i with the substitution $\Delta \rightarrow -\Delta$, $P \rightarrow -P$ and $\varphi \rightarrow -\varphi$. It is seen from Eq. (17) that $f_{x,y}$ are of the second order with respect to $\xi_S/d_W \equiv \sqrt{D/2\pi T}d_W$. $f_{0,z}$ also contain second order in ξ_S/d_W contributions, but they are not written here because they do not contribute to the anomalous current j_a .

Now substituting Eqs. (15)-(17) into the current Eq. (14) we obtain the following expression for the ordinary and anomalous currents:

$$j_a(x) = -\frac{8\pi TD\Gamma P}{e\rho_N} \sum_{\omega>0} \frac{f_z^2}{(\omega + \Gamma)^2} \left[(\partial_x\mathbf{a} \times \mathbf{a})_z + 2a_z(a_x^2 + a_y^2) \right] \quad (18)$$

$$j_o = -\frac{2\pi T}{e\rho_N} \sum_{\omega>0} [f_0^2 + f_z^2] \partial_x\varphi_0 \quad (19)$$

The ground state of the system is determined by the condition of zero total electric current $j_o + j_a = 0$. Using Eqs.(18), (19) we obtain the ground state that supports the gradient of superconducting phase

$$\partial_x\varphi_0 = \frac{j_a e\rho_N}{2\pi T \sum_{\omega>0} [f_0^2 + f_z^2]}. \quad (20)$$

Magnetic helix. Now we consider the special case of magnetic texture in the form of a helix. In this case $d\theta/dx = 0$ and $d\delta/dx = 2\pi\kappa/L$, where L is the spatial period of the helix and $\kappa = \pm 1$ determines its chirality.

$$\partial_x\varphi_0 = 8\pi^3 \cos\theta \sin^2\theta \frac{\kappa D\Gamma P}{L^3} \left(\frac{\sum_{\omega>0} f_z^2/(\omega + \Gamma)^2}{\sum_{\omega>0} [f_0^2 + f_z^2]} \right) \quad (21)$$

One can see that in this case the ground state of the superconductor corresponds to the helical state - the superconducting state with zero supercurrent and a constant phase gradient $\nabla\varphi = \partial_x\varphi_0\mathbf{e}_x$. Earlier the helical state has already been predicted for superconducting systems

with intrinsic spin-orbit coupling and under a uniform Zeeman field^{86–91}. Here we report that this state can be also realized in S/F spin-textured bilayers without an intrinsic spin-orbit coupling. It is also worth to note here that, while looking quite similar, this state is in sharp contrast to the famous FFLO state^{92,93}, where the direction of the phase gradient is not fixed by the exchange field. Here the direction of the phase gradient is strictly fixed by the magnetization texture.

Introducing the orthogonal vectors ($\hat{\mathbf{h}} \equiv \mathbf{h}/h$):

$$\mathbf{n}_\delta = -\partial_\delta \hat{\mathbf{h}} / \sin \theta \quad (22)$$

$$\mathbf{n}_\theta = -(\partial_x \mathbf{n}_\delta \partial_\theta \hat{\mathbf{h}}) \partial_\theta \hat{\mathbf{h}}, \quad (23)$$

which are also orthogonal to $\mathbf{h}$, we can also rewrite Eq. (21) in the form:

$$\partial_x \varphi_0 = 4\pi^2 \chi_{int} P \sin^2 \theta \frac{D\Gamma}{L^2} \left(\frac{\sum_{\omega>0} f_z^2 / (\omega + \Gamma)^2}{\sum_{\omega>0} [f_0^2 + f_z^2]} \right) \quad (24)$$

where $\chi_{int} = \hat{\mathbf{h}}(\mathbf{n}_\theta \times \mathbf{n}_\delta) = (2\pi\kappa/L) \cos \theta$ is the invariant, describing the internal chirality of the helix. $P\chi_{int}$ is the same chiral invariant which was introduced in Ref. 51 to describe the anomalous Josephson effect in S/F/S junctions with a helix magnetic interlayer. It illustrates the universality of the chiral nature of the inverse magnetoelectric effect in superconducting hybrids with textured ferromagnets.

Eqs. (21) and (24) describe the helical state in the limit of slow helices $\xi_S/L \ll 1$. The dependence of $\partial_x \varphi_0$ on the inverse helix period in more general case of arbitrary L (still larger than the atomic scales) is presented in Fig. 2. It is seen that that maximal values of the phase gradient are reached for helices with the period $L \sim 0.15\xi_S$. Fig. 2(b) also demonstrates that the stronger the spin-dependent part of the depairing $P\Gamma$ the larger the phase gradient of the helical state. In fact, the dependence $\partial_x \varphi_0(\Gamma)$ is nonmonotonous and $|\partial_x \varphi_0|$ is strongly reduced at $\Gamma > T_c$. But this part of the curve $\partial_x \varphi_0(\Gamma)$ is not plotted because the superconductivity by itself is already suppressed at such strong depairing factors.

Domain wall. Further we consider a domain wall case. For definiteness we focus on the head-to-head domain wall with $\theta(x) \rightarrow 0, \pi$ at $x \rightarrow \mp\infty$.

If the wall is coplanar, what is equivalent to the condition $\partial_x \delta = 0$, then $j_a = 0$, as it can be easily seen from Eq. (18) and the definition of vector $\mathbf{a}$. In this case the magnetoelectric effect is absent and the ground state of the superconductor is a homogeneous state with a spatially constant phase.

If the DW is noncoplanar due to some reasons, that is $\partial_x \delta \neq 0$, $j_a \neq 0$ and between any points x_1 and x_2 of the superconductor in the ground there is a phase difference, which is to be calculated as $\int_{x_1}^{x_2} \partial_x \varphi_0(x) dx$, where $\partial_x \varphi_0$ is defined by Eq. (20). It is obvious that this phase difference is zero far from the DW, but is finite if the

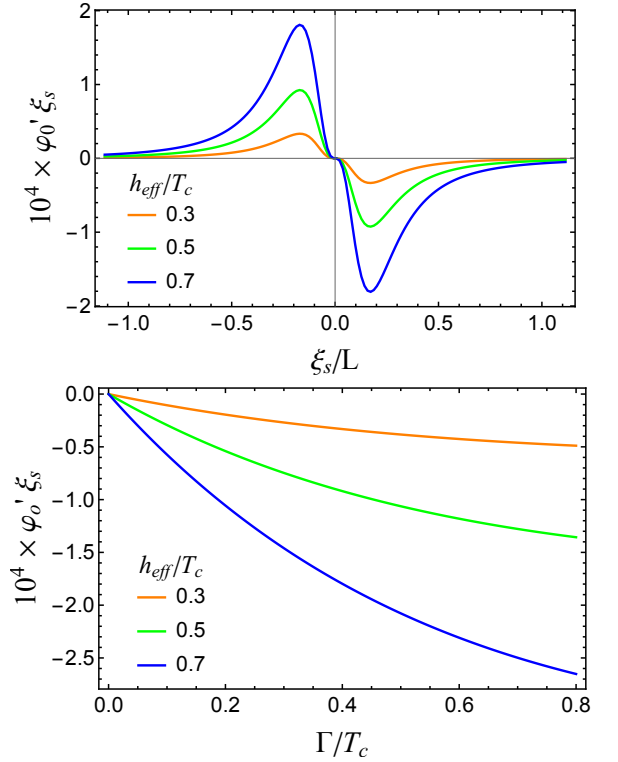


FIG. 2. (a) $\partial_x \varphi_0$ as a function of the inverse helix period L^{-1} for different h_{eff} . $\Gamma = 0.4T_c$. (b) $\partial_x \varphi_0$ as a function of Γ for different h_{eff} . $2\pi\xi_S/L = 1$. For the both panels $P = 0.5$, $\theta = \pi/3$.

DW is located inside the region between x_1 and x_2 . The noncoplanarity of the wall can be caused by different reasons. For example, it can be induced by the supercurrent moving the DW, or it can be just due to the contact between the ferromagnet and the superconductor, because in this case it is energetically more favorable to disturb the initial wall texture in order to reduce stray fields penetrating the superconductor. In any of the described cases the resulting magnetization structure and the ground state phase difference can be calculated, but this problem is beyond the scope of the present work. Instead, here we consider the case of the externally induced noncoplanarity in the system and demonstrate that in this case the resulting ground state phase difference is also governed by a chiral invariant.

The simplest model system where the external (not caused by the internal chirality of the ferromagnet texture) chiral invariant takes place is sketched in Fig. 1(b). It is assumed that the wall is coplanar, but the polarization $\mathbf{P}$ of the spin-filtering interface is not fully aligned with the ferromagnet magnetization.

In this case we start from Eq. (7) valid for any mutual orientation of the exchange field and the polarization term. In order to obtain nonzero anomalous current j_a for the considered case of external chirality it is enough to find the anomalous Green's functions up to the zero

order in ξ_S/d_W . Then the linearized Usadel equations for the anomalous Green's function take the form:

$$\frac{D}{2}\partial_x^2 \mathbf{f} - (\omega + \Gamma)\mathbf{f} + ih_{eff}f_0\mathbf{e}_z - i\Gamma[\tilde{\mathbf{P}} \times \mathbf{f}] = 0, \quad (25)$$

$$\frac{D}{2}\partial_x^2 f_0 - (\omega + \Gamma)f_0 + ih_{eff}f_z + i\Delta = 0. \quad (26)$$

The solution of these equations up to the first order with respect to $|\tilde{\mathbf{P}}(x)|\Gamma/(\omega + \Gamma)$ can be written in the compact form:

$$\mathbf{f}_\perp = \frac{-i\Gamma f_z}{(\omega + \Gamma)}[\tilde{\mathbf{P}} \times \mathbf{e}_z], \quad (27)$$

where $\mathbf{f}_\perp = (f_x, f_y, 0)$ is the component of the triplet anomalous Green's function $\mathbf{f}$ in plane perpendicular to the local quantization axis. The components f_z and f_0 are expressed by Eqs. (16) and (15), respectively.

For the coplanar domain wall vector $\mathbf{a} = (1/2)(d\theta/dx)\mathbf{e}_x$ and the anomalous current can be expressed as

$$j_a(x) = \frac{4\pi T\Gamma}{e\rho_N} \sum_{\omega>0} \frac{f_z^2}{\omega + \Gamma} (\mathbf{P}[\partial_x \hat{\mathbf{h}} \times \hat{\mathbf{h}}]). \quad (28)$$

The corresponding ground state phase gradient can be found according to Eq. (20). The total phase difference acquired in the superconductor due to the DW presence can be found as $\Delta\varphi_0 = \int_{-\infty}^{\infty} \partial_x \varphi_0 dx$ and takes the form:

$$\Delta\varphi_0 = 2\pi\Gamma|P_\perp|\chi_{ex} \frac{\sum_{\omega>0} f_z^2/(\omega + \Gamma)}{\sum_{\omega>0} [f_0^2 + f_z^2]}, \quad (29)$$

where $\chi_{ex} = \text{sgn}[\mathbf{P}(\partial_x \hat{\mathbf{h}} \times \hat{\mathbf{h}})]$, $P_\perp$ is the component of the polarization vector $\mathbf{P}$ perpendicular to the DW plane. We see that the ground state phase difference acquired by the superconductor due to the presence of the DW in the ferromagnet is controlled by the external chirality invariant χ_{ex} , which is only nonzero if the polarization $\mathbf{P}$ of the spin-filtering interface has the component perpendicular to the wall plane.

Eq. (29) gives the ground state phase difference at the domain wall in the limit of wide DW $d_W \gg \xi_S$. Beyond this limit our analytical treatment is inapplicable, but the result of the numerical calculation is presented in Fig. 3. It is seen that the maximal phase difference is acquired at wide walls with $d_W \gg \xi_S$, when $\Delta\varphi_0$ tends to the analytical answer expressed by Eq. (29).

In principle, there can be also a contribution to the anomalous current from the vector potential of the stray fields generated by the DW. However, this contribution depends on y -coordinate and should be zero after averaging over it. Therefore, the averaged over the y -coordinate phase difference is still given by considered magnetoelectric effect even if the stray fields are taken into account.

In general the discussed above phase difference arising in a superconductor due to the magnetoelectric effect

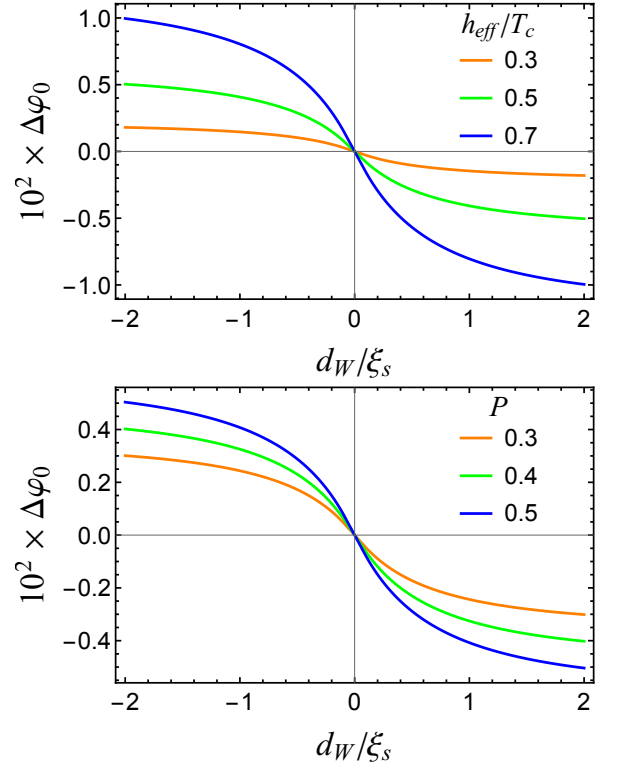


FIG. 3. Phase difference acquired at the DW in case of external chirality (see text) as a function of d_W^{-1} . (a) Different curves correspond to different h_{eff} , $P = 0.5$. (b) Different curves correspond to different P , $h_{eff} = 0.5T_c$. For the both panels $\Gamma = 0.4T_c$.

provides the connection between the the magnetic texture and the condensate phase. In particular, it opens a way to detect electrically time-dependent textures (for example, moving domain wall and other magnetic defects) via the relation $V = (\hbar/2e)\partial_t \Delta\varphi$. It is interesting that such an electrical detection resolves not only the defect movement, but also its chirality. The detailed investigation of this problem is beyond the framework of this work and we postpone it for a future study.

IV. DIRECT MAGNETOELECTRIC EFFECT IN S/F BILAYERS AND THE RELATED TORQUE

Here we consider the direct magnetoelectric effect in S/F bilayers, that is the generation of an equilibrium spin polarization in response to a supercurrent. The induced polarization is found to be perpendicular to the ferromagnet magnetization and, therefore, gives rise to a torque acting on the ferromagnet texture $\mathbf{M}(\mathbf{r})$. Further we study the torque in details.

Our first goal is to find the supercurrent-induced spin polarization in the superconductor $\mathbf{m} = -2\mu_B \mathbf{s}$, where $\mathbf{s}$ is an electron spin. In terms of the linearized quasi-classical Green's function in the fixed spin basis it can be

calculated as follows:

$$\mathbf{m} = \frac{-i\pi\mu_B N_F T}{2} \sum_{\omega>0} \text{Tr}_2 [\boldsymbol{\sigma}(\hat{f}\hat{f} + \hat{f}\hat{f})]. \quad (30)$$

If we define the spin vectors $\boldsymbol{\alpha}^i$ for $i = x, y, z$ as $\boldsymbol{\alpha}^i \boldsymbol{\sigma} = U^\dagger \sigma_i U$, then in terms of the quasiclassical Green's function in the local spin basis Eq. (30) can be written as follows:

$$\mathbf{m}_i = -2i\pi\mu_B N_F T \sum_{\omega>0} \boldsymbol{\alpha}^i (f_0 \tilde{\mathbf{f}} + \tilde{f}_0 \mathbf{f}). \quad (31)$$

In order to calculate the induced electron magnetization according to Eq. (31) we have to find the anomalous Green's functions in the local spin basis up to the first order in the applied supercurrent, or in other words, up to first order in the superconducting phase gradient $\partial_x \varphi$. We start from the linearized Usadel equations, valid for the case of an arbitrary directed polarization of the spin-filtering interface:

$$\frac{D}{2} \partial_x^2 \mathbf{f} - (\omega + \Gamma + 2Da^2) \mathbf{f} - D([\partial_x \mathbf{a} \times \mathbf{f}] + 2[\mathbf{a} \times \partial_x \mathbf{f}]) - i\Gamma[\tilde{\mathbf{P}} \times \mathbf{f}] + 2D(\mathbf{a}\mathbf{f})\mathbf{a} = -ih_{eff} f_0 \mathbf{e}_z, \quad (32)$$

$$\frac{D}{2} \partial_x^2 f_0 + ih_{eff} f_z + i\Delta(x) = (\omega + \Gamma) f_0. \quad (33)$$

Assuming that $\Delta(x) = \Delta e^{i\varphi(x)}$ and performing the transformation $\hat{f}_l = \hat{f}_h e^{i\varphi(x)}$, up to the first order in $\partial_x \varphi$ and up to the first order in ξ_S/d_W Eqs. (32)-(33) take the form

$$\frac{D}{2} \partial_x^2 \mathbf{f}_h - (\omega + \Gamma) \mathbf{f}_h - 2iD\partial_x \varphi [\mathbf{a} \times \mathbf{f}_h] - i\Gamma[\tilde{\mathbf{P}} \times \mathbf{f}_h] = -ih_{eff} f_{h0} \mathbf{e}_z, \quad (34)$$

$$\frac{D}{2} \partial_x^2 f_{h0} - (\omega + \Gamma) f_{h0} + i\Delta = -ih_{eff} f_{hz} \quad (35)$$

The solution of these equations up to the first order in $P\Gamma/(\omega + \Gamma)$ takes the form

$$f_{hz} = f_z \left(1 + \frac{4D\partial_x \varphi \Gamma}{(\omega + \Gamma)^2 + h_{eff}^2} \mathbf{a}_\perp \tilde{\mathbf{P}}_\perp \right), \quad (36)$$

$$f_{h0} = f_0 + \frac{ih_{eff}}{(\omega + \Gamma)} \frac{4D\partial_x \varphi \Gamma f_z}{(\omega + \Gamma)^2 + h_{eff}^2} \mathbf{a}_\perp \tilde{\mathbf{P}}_\perp, \quad (37)$$

$$\mathbf{f}_{h\perp} = \mathbf{f}_\perp - \frac{2i\partial_x \varphi D[\mathbf{a} \times \mathbf{e}_z] f_z}{\omega + \Gamma} - \quad (38)$$

$$\frac{2\partial_x \varphi D\Gamma f_z}{(\omega + \Gamma)^2} [\mathbf{a}_\perp \tilde{\mathbf{P}}_\perp + \tilde{\mathbf{P}}_z \mathbf{a}_\perp],$$

where f_z , f_0 and $\mathbf{f}_\perp$ are defined by Eqs. (16), (15) and (27), respectively. $\tilde{\mathbf{P}}_\perp$ is the component of the vector $\tilde{\mathbf{P}}$, which is perpendicular to the local direction of the ferromagnet magnetization. $\mathbf{a}_\perp$ is defined in the same way. The anomalous Green's function $\hat{f}_h$ can be obtained

from Eqs. (36)-(38) with the substitution $\Delta \rightarrow -\Delta$, $P \rightarrow -P$ and $\partial_x \varphi \rightarrow -\partial_x \varphi$.

Substituting the anomalous Green's functions into Eq. (31) we obtain the following result for the supercurrent-induced electron magnetization:

$$\mathbf{m} = 4\pi\mu_B N_F T D \partial_x \varphi \sum_{\omega>0} \frac{f_z}{(\omega + \Gamma)[(\omega + \Gamma)^2 + h_{eff}^2]} \times \Gamma \Delta \left\{ (\partial_x \hat{\mathbf{h}} \times \hat{\mathbf{h}})(\mathbf{P}\hat{\mathbf{h}}) - \delta' \cos \theta \tilde{\mathbf{P}}_\perp \right\}. \quad (39)$$

Here we have only written the magnetization component, which is perpendicular to the local direction of the ferromagnet $\hat{\mathbf{h}}$ because it is this component that gives rise to a torque $\mathbf{N}$ acting on the ferromagnet magnetization. The torque can be calculated as follows:

$$\mathbf{N} = 2h\beta(\hat{\mathbf{h}} \times \mathbf{m}), \quad (40)$$

where $\beta = h_{eff}/h < 1$ is the dimensionless coefficient between the actual exchange field of the ferromagnet h and the effective exchange field h_{eff} , which is induced in the superconductor due to the magnetic proximity effect. Substituting $\mathbf{m}$ from Eq. (39) the torque can be written as

$$\mathbf{N} = b_j \partial_x \hat{\mathbf{h}} + c_j \chi_{int}(\tilde{\mathbf{P}}_\perp \times \hat{\mathbf{h}}), \quad (41)$$

$$c_j = \pi\mu_B N_F T D \partial_x \varphi \sum_{\omega>0} \frac{8\beta h \Gamma \Delta f_z}{(\omega + \Gamma)[(\omega + \Gamma)^2 + h_{eff}^2]}, \quad (42)$$

$$b_j = c_j(\mathbf{P}\hat{\mathbf{h}}), \quad (43)$$

Here χ_{int} is the local internal chirality of the ferromagnet texture, defined in the same way as for the magnetic helix case:

$$\chi_{int} = \delta' \cos \theta = \hat{\mathbf{h}}(\mathbf{n}_\theta \times \mathbf{n}_\delta), \quad (44)$$

with $\mathbf{n}_\delta$ and $\mathbf{n}_\theta$ defined by Eqs. (22)-(23). The expression for coefficient c_j (42) can be rewritten in terms of the supercurrent flowing via the superconductor:

$$c_j = -2\beta \Gamma \Delta \mu_B h j \times \sum_{\omega>0} \frac{f_z}{(\omega + \Gamma)[(\omega + \Gamma)^2 + h_{eff}^2]} \bigg/ \sum_{\omega>0} (f_{h0}^2 + f_{hz}^2), \quad (45)$$

where up to the leading order in ξ_S/d_W we have neglected the anomalous current j_a .

The first term in Eq. (41) represents the adiabatic spin transfer torque (STT). Typically the adiabatic spin transfer torque is originated from the transfer of the angular momentum from the current-carrying electrons to the ferromagnet magnetization and, consequently, coefficient b_j is proportional not only to the electric current j , but also the degree of its spin polarization.

Here the microscopic origin of the adiabatic STT is different. At first let us consider the case when $\mathbf{P}$ is

aligned with the ferromagnet magnetization. Then $\tilde{\mathbf{P}}_{\perp} = 0$ and the adiabatic STT is the only contribution to the torque in the system. It can be demonstrated that in this case the spin current through the system is zero to the considered accuracy. Therefore, the electric current is not spin polarized and the torque is not connected to the derivative of the spin current, that is to the spin transfer from the current-carrying electrons to the magnetization. Its mechanism is connected to the creation of current-induced spin-resolved DOS in the superconductor in the region contacted to the textured area of the ferromagnet. Therefore, it is specific only for superconducting systems and can also be relevant as for hybrid superconducting systems with ferromagnetic metals, so as for hybrids with ferromagnetic insulators.

If $\mathbf{P}$ is not fully aligned with the ferromagnet magnetization, the other part of the torque, expressed by the second term in Eq. (41) can appear. In general, it has components as along the direction $\partial_x \hat{\mathbf{h}}$, so as along the perpendicular direction $\hat{\mathbf{h}} \times \partial_x \hat{\mathbf{h}}$. But it cannot be included neither to the adiabatic STT, nor to the non-adiabatic STT, because it is proportional to the local internal chirality of the structure χ_{int} and, consequently, vanishes for coplanar ferromagnetic textures.

V. CONCLUSIONS

We have studied the direct and inverse magnetoelectric effects in thin film S/F bilayers with spin-textured ferromagnets. The generalized Usadel equation, which allows for description of the magnetoelectric effects, is formu-

lated. The inverse magnetoelectric effect leads to the formation of the phase-inhomogeneous ground state in the superconducting film due to the exchange interaction of spin-triplet Cooper pairs with non-coplanar magnetic texture. This effect can be used for electrical detection of DW motion.

The direct magnetoelectric effect induces a stationary spin polarization of the superconducting condensate in the presence of the applied supercurrent and the non-coplanar spin texture. The direction of induced Cooper pair spin is non-collinear with the local magnetization in the adjacent ferromagnetic layer. Therefore exchange interaction of the induced electron spin and the ordered magnetic moments acts as a spin torque on the ferromagnet magnetization. This torque consists of two parts. The first one is similar to usual adiabatic spin-transfer torque, and the second one is connected to the local chirality of the magnetic texture. The chirality-sensitive term in the spin torque can be mediated only by the spin-triplet superconducting correlations since it is generically absent in the normal state. The found superconducting spin torque in S/F bilayers can be used in spintronics for the low-dissipative electric current-controlled manipulation with positions of domain walls and magnetic skyrmions.

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